

Assignment – 3: AI and Biotechnology/Bioinformatics

Summary Report –

Assignment – 3 Exploratory Data Analysis & Descriptor/Fingerprint Calculation

Target Protein: BCL2

The dataset used in this analysis was curated in Assignment 2 and includes **ChEMBL IDs, SMILES structures, pIC50 values, and bioactivity class (active/inactive)** for compounds targeting BCL2.

Exploratory Data Analysis (EDA)

Basic statistical analysis was performed on the **pIC50 values** to understand the bioactivity distribution of the compounds.

pIC50 Statistics

- **Mean:** ~7.04
- **Minimum:** 4.93
- **Maximum:** 8.35
- **Standard Deviation:** ~1.20

A **histogram of pIC50 values** was generated to visualize the distribution of compound activity. The plot shows that most compounds are clustered around **higher pIC50 values**, indicating moderate to strong inhibitory activity.

A **bar plot of bioactivity classes** was also generated to compare **active vs inactive compounds**. The visualization shows that **active compounds are more prevalent in the dataset**, suggesting enrichment of biologically relevant inhibitors.

To evaluate descriptor significance, a **Mann–Whitney U statistical test** was performed.

Significant descriptors ($p < 0.05$):

- Molecular Weight (MW)
- LogP
- Number of Hydrogen Bond Donors
- Number of Hydrogen Bond Acceptors

These descriptors show statistically significant differences between active and inactive compounds.

Lipinski and 2D Descriptor Statistics

Lipinski molecular descriptors were calculated to evaluate drug-likeness.

Approximate descriptor ranges:

- **Molecular Weight (MW):** 442 – 841
- **LogP:** 4.92 – 8.73
- **Hydrogen Bond Donors:** 0 – 3
- **Hydrogen Bond Acceptors:** 0 – 10

These descriptors provide insights into **molecular size, lipophilicity, and hydrogen bonding capacity**, which influence drug absorption and interaction with the **BCL2 protein target**.

Fingerprint Calculation

Fingerprint type generated:

PubChem fingerprints

Number of fingerprint features:

881 binary bits

Tool used:

PaDEL Descriptor / PaDELpy

Reason for selecting this method:

PaDEL is a widely used cheminformatics tool that efficiently calculates **molecular descriptors and standardized fingerprints from SMILES structures**. PubChem fingerprints provide **881 structural features**, which are useful for **chemical similarity analysis, QSAR modeling, and machine learning applications**.

✓ Outputs Generated

- pIC50 histogram
- Bioactivity class bar plot
- Descriptor boxplots (MW, LogP, H-donors, H-acceptors)
- Lipinski descriptor dataset
- Mann-Whitney statistical significance results

These analyses provide a **clear understanding of chemical properties and bioactivity trends of compounds targeting the BCL2 protein.**